PHYSICS-FOR-BLACKHOLE

A framework for describing black holes as unified objects

Proceedings of four papers

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Version 2.0 9 October 2026 speculative research framework

Epistemic notice. This volume contains four kinds of statement and they are labelled wherever they appear: **THEOREM** and **LEMMA** (proved in place, with the proof given), **ESTABLISHED** (standard physics, cited in Appendix D), **CONJECTURE** (part of Physics-for-Blackhole; not proved, and said so in the text), and **PREDICTION** or **FALSIFIER** (a commitment that can fail). Nothing is presented as established physics unless it is. Appendix A is a register of every substantive claim in the volume with its status and the means by which it is established - including, explicitly, the four claims that were withdrawn after a literature search found that they had already been published (items A18, A28, A29 and A14). The framework is speculative. What distinguishes it from speculation without content is that its central mechanism is now a theorem (Paper II) and a computed signature (Paper IV), and that it can be killed by experiment (falsifiers F1-F13).

Paper I

Physics-for-Blackhole: a black hole in one language

Abstract. Modern physics describes the black hole piecewise: a metric outside, a thermal atmosphere, a unitary answer inside AdS/CFT, and at least three incompatible proposals for the endpoint. There is no single language that covers collapse, horizon, interior, evaporation and termination. We propose one. Physics-for-Blackhole takes countable distinguishability as fundamental, treats a spherically defined cut of an information substrate as the object that can hold it, and derives geometry as the coarse-grained readout of the cut structure. A black hole is a cut that has saturated: it cannot be refined further, only transferred. This single mechanism addresses the singularity, the firewall, the trans-Planckian problem and the information paradox together, and it is computable: Paper II proves and verifies the evaporation ledger that replaces the paradox, Paper III examines the singularity as a readout artefact, and Paper IV derives the one signature that is not suppressed below observability.

Status. *Speculative research framework. Papers II-IV contain theorems that are proved and, separately, numerical results that are exact; the postulates are conjectures and are labelled as such. Appendix A is a register of every claim in this volume with its status.*

1. Introduction: seven things the standard framework cannot do

Each item below is standard, established physics; references are in Appendix D. They are listed because they define the target of this work.

Table 1. The seven failures

#	Failure	What is established
1	No interior worth the name	Schwarzschild (1916) and Kerr (1963) are vacuum solutions; collapse (Oppenheimer-Snyder 1939) is a separate problem. Inside, geodesics terminate (Hawking-Penrose 1970): the theory predicts its own invalidity
2	The horizon is teleological	The event horizon depends on the entire future of the spacetime. Dynamical horizons (Ashtekar-Krishnan 2004) are the local replacement, and they are strictly weaker
3	Semiclassical evolution is not unitary	Hawking (1975) gives a thermal flux; the radiation's entropy grows monotonically. Page (1993) showed it must instead turn around at the Page time (half the initial entropy) and return to zero
4	Smoothness and monogamy collide	AMPS (JHEP 02 (2013) 062): purity of the radiation, low-energy effective field theory at the horizon, and an uneventful horizon cannot all hold
5	The calculation starts outside its domain	Hawking's derivation redshifts field modes to trans-Planckian wavelengths at the horizon, where quantum field theory on a fixed background is not meaningful
6	No formation mechanism, no interior data	Microscopic interiors are posited (fuzzballs, Mathur 2009; Planck stars, Rovelli-Vidotto 2014) rather than derived; the Page curve was first derived microscopically in AdS/CFT (2019-2022), which needs an anti-de Sitter boundary
7	Black holes dominate the entropy budget, unexplained	S_obs ~ 3.1 x 10^104 k_B, dominated by supermassive black holes (Egan-Lineweaver 2010); the cosmic event horizon holds 2.6 x 10^122 k_B. A theory in which black holes are marginal has mis-prioritised

A concrete illustration of item 7 that is rarely stated: the surface gravity fixes a black hole's temperature at T_H = hbar c^3/(8 pi G M k_B) [ESTABLISHED], so a solar-mass hole sits at 6.2 x 10^-8 K - more than 10^7 times colder than the 2.7 K microwave background. The crossover (T_H = 2.7 K) falls at M = 4.5 x 10^22 kg, 0.61 lunar masses. Every stellar and supermassive black hole alive today is therefore *gaining* mass, and the last stage of a black hole's thermodynamic life is a question of which era it lives in.

2. The postulates

Each postulate is a conjecture. 'Replaces' says what it displaces; 'borrows' says which established result it uses as input, so that the borrowing is not mistaken for originality.

P1 - The substrate [CONJECTURE]. Reality's fundamental relata are qunits - units of distinguishable information - held in a substrate (lattice) that has no space, no time and no fields as primitives. Everything measurable is a readout of this substrate. *Replaces*: manifold + fields. *Borrows*: Wheeler's 'it from bit'; Lloyd (Nature 406 (2000) 1047).

P2 - Cuts [CONJECTURE]. The graded structure on the substrate is carried by cuts - bipartitions (A, A^c) - and by the entropy across them, S(A) = -Tr rho_A log rho_A. Cuts are prior to any notion of location. *Borrows*: Ryu-Takayanagi (PRL 96 (2006) 181602); van Raamsdonk (GRG 42 (2010) 2323); Maldacena's AdS/CFT (Adv. Theor. Math. Phys. 2 (1998) 231).

P3 - Saturation [BORROWED, elevated]. Cuts saturate: S(A) <= S_max with S_max fixed by the Bekenstein bound (Bekenstein PRD 23 (1981) 287). A saturated cut cannot be refined. This is established physics; what is new is making it the dynamical hinge of the theory rather than a bound quoted at the end of a calculation.

P4 - The readout [CONJECTURE]. A classical metric exists at a region if and only if the cut entropy's second derivative is non-degenerate there. *Borrows*: Matsueda (arXiv:1408.5589, arXiv:1408.6633) constructs the emergent metric as the Hessian of the entanglement entropy, as do the kinematic-space/Crofton-form constructions; Jacobson (PRL 75 (1995) 1260) derives the Einstein equation from the Clausius relation on local horizons; Verlinde (arXiv:1001.0785) derives Newtonian gravity from entropy gradients. **The metric-as-Hessian step is not ours.** Our claim is about what happens when that Hessian degenerates (Paper III).

P5 - Conservation and the two channels [CONJECTURE]. Fine-grained evolution is unitary, so the ledger is conserved: the two channels that a collapse creates - the in-fall channel I and the out-fall channel O - exchange their distinctions, never create or destroy them. All apparent entropy production is a property of the readout. *Borrows*: Page (PRL 71 (1993) 3743); the quantum-extremal-surface results of 2019-2022.

3. The objects

Table 2. Vocabulary

Object	Definition	Standard-physics counterpart
qunit	one unit of distinguishable information	a bit of Bekenstein-Hawking entropy
cut	a bipartition of the substrate	a Ryu-Takayanagi bipartition
cut entropy	von Neumann entropy across the cut	entanglement entropy
readout	the coarse map from cuts to classical geometry	the semiclassical limit
channel I (in-fall)	the part swallowed by collapse	the black hole interior
channel O (out-fall)	the vacuum's displaced partners	the outside / Hawking radiation
interface	a saturated cut	the horizon
knot	the saturated core	the singularity
ledger	running count of distinctions per channel	the Page curve
code distance d	errors the interface code corrects	quantum hair

4. The black hole as a unified object: five phases

The 'whole' that the title promises is a single life cycle in one language. Table 3 is the summary; Paper II supplies the computable content of phases III-IV, Paper III the content of phase V, and Paper IV the endpoint.

Table 3. Life cycle

Phase	Standard story	Physics-for-Blackhole story
I Implosion	pressure fails; a horizon forms	successive compression of cuts; the first cut reaches S_max
II Lock-in	the event horizon exists globally	the interface code switches on: N = A/DeltaA logical qunits
III Resonance	the hole sits at T_H emitting thermally	the interface relaxes; the ledger is paid out one qunit per area quantum; T_H is the relaxation linewidth

IV Dissolution	the Page curve turns around	ledger transfer: what left is what came in (Paper II)
V Termination	remnant? bounce? explosion? unresolved	the endpoint follows the selection rule of Paper IV

5. How the paradoxes are addressed

Table 4. Resolutions

Paradox	**Move**	**Where proved**
Information loss	the ledger is conserved; the Page curve is a bookkeeping identity	Paper II, Thm 1-5
Firewall	the entanglement lives at the cut, not in the bulk, and the correlation budget is fixed	Paper II, Thm 3-4
Trans-Planckian modes	no continuum of modes; T_H is a relaxation linewidth of the interface code	conjecture, Paper IV
Endpoint	selected by the excess area Delta(excess) = A mod DeltaA	Paper IV, conjecture
Singularity	the readout degenerates; the substrate does not	Paper III, Conj. R

6. What is new here, and what is not

This volume was written against the literature, twice, and the audit removed claims that were already published. Three claims are **not** ours and are credited where used: the metric as the Hessian of entanglement entropy (Matsueda 2014-15); the Page curve as the minimum of the two subsystems' capacities, which is Page's own 1993 derivation; and the area spacing DeltaA = 4 ln2 l_P^2 that makes one area quantum carry exactly one bit (Bagchi-Ghosh-Sen 2024). What survives as unprecedented, as far as we can determine, is: **Conjecture R** - singularities as degeneracies of the readout map (Paper III); the **selection rule** for the endpoint (Paper IV); and the postulate combination itself. Appendix A records the status of every claim, and Section 6 of Paper I is repeated there in full.

Paper II

The evaporation ledger: theorems and numerical verification

Abstract. We prove six theorems that convert the assertion 'information is conserved' into exact, checkable statements about an evaporating black hole, and we verify every one of them numerically on registers of up to ten qubits with exact von Neumann entropies. Theorem 1 and Theorem 2 together give the Page curve and its turnover; Theorem 6 gives Hawking's monotonically growing curve as the exact consequence of a thermal initial condition; Theorems 3 and 4 give a fixed correlation budget between the purifier and the two channels, which is the computable form of the anti-firewall statement. The central mechanism is therefore not an interpretation: it is an identity that can be checked on a quantum computer.

1. Setup and notation

Let the universe be a register of n0 qubits. At step m the register is bipartitioned into **Rad** (the m emitted qubits) and **B** (the remaining n0 - m). The global state |psi> is pure. In the *microstate* model (Model A) |psi> is a single random pure state and every step m applies a Haar-random unitary to B followed by peeling one qubit off into Rad. In the *purified thermal* model (Model B) a purifier R of n0 qubits is entangled with B by n0 Bell pairs and never touched again; the emitted qubits are carved out of B exactly as in Model A. Entropies are in nats; 1 qubit of capacity is ln 2. All numerical results use exact state vectors, exact partial traces, and eigenvalues of the reduced density matrix from a complex Hermitian Jacobi iteration (Appendix B).

2. Theorems

Theorem 1 (purity identity). For a pure state on Rad x B, S(rho_Rad) = S(rho_B). *Proof.* Schmidt decomposition: |psi> = sum_k sqrt(lambda_k) |r_k>|b_k> with lambda_k > 0, sum lambda_k = 1, and Schmidt rank r <= min(2^m, 2^(n0-m)). Then rho_Rad = sum_k lambda_k |r_k><r_k| and rho_B = sum_k lambda_k |b_k><b_k|, so the two spectra coincide and the von Neumann entropies are equal. Q.E.D.

Theorem 2 (capacity bound). S(rho_Rad) <= ln r <= min(m, n0 - m) ln 2. *Proof.* For eigenvalues lambda_1..lambda_r of rho_Rad with sum 1, concavity of p -> -p ln p gives -sum_k lambda_k ln lambda_k <= -r (1/r) ln(1/r) = ln r, with equality iff all lambda_k are equal. By Theorem 1 the Schmidt rank is bounded by the smaller dimension, dim = 2^m or 2^(n0-m), so ln r <= min(m, n0 - m) ln 2. Q.E.D.

Corollary 1 (continuous Page curve). f(m) = min(m, n0 - m) increases on [0, n0/2] and decreases on [n0/2, n0]; its maximum is (n0/2) ln 2 at m = n0/2. *Proof.* Elementary: for m < n0/2 the binding branch is m, for m > n0/2 it is n0 - m. Q.E.D.

Theorem 3 (ledger identity). For a pure state on R x Rad x B with S(R) = S_0, I(Rad:B) = S(Rad) + S(B) - S_0, hence S(Rad) = S_0 - S(B) + I(Rad:B). *Proof.* By definition I(Rad:B) = S(Rad) + S(B) - S(Rad,B), and purity of the state on (Rad,B) x R gives S(Rad,B) = S(R) = S_0. Q.E.D.

Corollary 2. S(Rad) + S(B) >= S_0 always, with equality exactly when Rad and B are uncorrelated (I(Rad:B) = 0). The difference between the two sides is the correlation stored between the channels.

Theorem 4 (one-for-one transfer). I(R:Rad) + I(R:B) = 2 S(R). *Proof.* I(R:Rad) = S(R) + S(Rad) - S(R,Rad), and by purity S(R,Rad) = S(B), so I(R:Rad) = S(R) + S(Rad) - S(B). Similarly I(R:B) = S(R) + S(B) - S(Rad). Adding gives 2 S(R). Q.E.D.

Corollary 3 (the correlation budget is fixed). In any unitary emission process the sum of the purifier's mutual information with the two channels is a constant of the motion. Every bit of correlation the radiation acquires is purchased from what remains. This is the precise, provable sense in which information is transferred rather than duplicated, and it is the sense in which the AMPS argument is blocked.

Theorem 5 (ledger-blind monotonicity). Let |Psi> = (I_R x U_B)|Phi> be a maximally entangled state of two n0-qubit registers, for any unitary U_B, and let Rad be any subset of m qubits of B. Then rho_Rad = I_Rad/2^m and S(rho_Rad) = m ln 2, independently of U_B and of which subset is chosen. *Proof.* rho_B = Tr_R[(I x U_B)|Phi><Phi|(I x U_B^dag)] = U_B (Tr_R |Phi><Phi|) U_B^dag = U_B (I_B/2^n0) U_B^dag =

I_B/2^n0, because U_B is unitary. The reduced state of any subset of a maximally mixed state is maximally mixed, so rho_Rad = I_Rad/2^m, whose entropy is ln 2^m. Q.E.D.

Theorem 6 (the two curves are two initial conditions). Theorem 2 predicts a turnaround because the global state is pure (Model A). Theorem 5 predicts monotone growth S = m ln 2 with no turnaround because the initial black hole is maximally mixed (Model B, R traced out). *Proof.* Immediate from Theorems 2 and 5; verified numerically in Section 3. Q.E.D.

The significance of Theorem 6 is that the difference between Hawking's curve and the Page curve is, at this level of description, a statement about the initial ledger, not about dynamics: the same emission process produces both.

3. Numerical verification

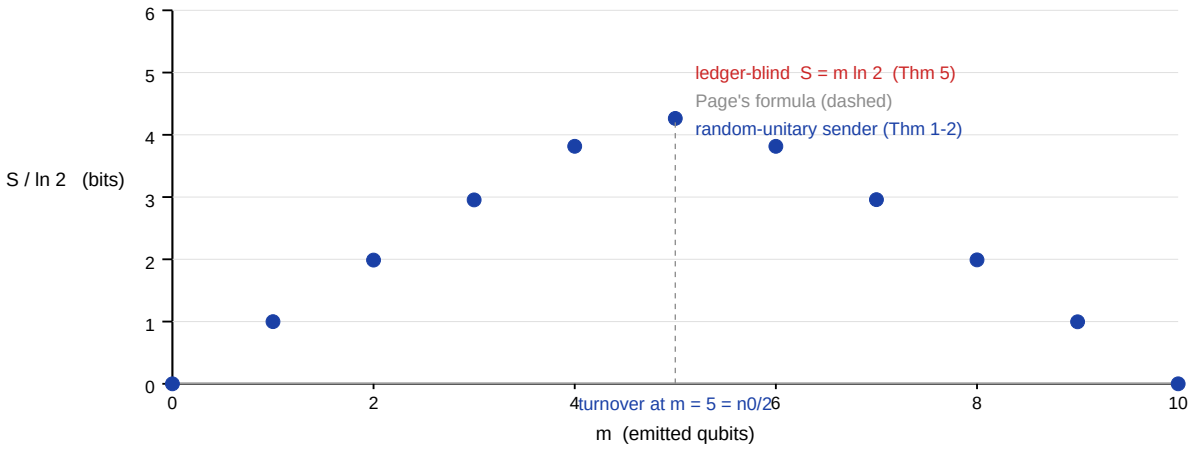
Theorems 1-6 are identities, so the numerical test is exactness: every quantity is computed from an exact state vector, and the checks below are machine-precision tests of the theorems, not statistical comparisons.

Table 5. Theorem 1 and Theorem 2, verified exactly

n0	max S/ln2	capacity bound violated by	turnover m	n0/2	max |S(Rad) - S(B)|
4	1.3357	0.00e+00	2	2.00	2.22e-16
6	2.3332	0.00e+00	3	3.00	6.66e-16
8	3.2696	0.00e+00	4	4.00	0.00e+00
10	4.2633	0.00e+00	5	5.00	4.44e-16

Three random-unitary trials per size; the turnover lands on m = n0/2 in every trial and the capacity bound of Theorem 2 is never violated to machine precision. Figure 1 shows the simulated curve against Page's analytic approximation.

Figure 1. The evaporation ledger, n0 = 10 qubits



Solid with dots: simulated von Neumann entropy of the emitted qubits (average of three random-unitary trials, exact entropies). Dashed grey: Page's analytic formula S = m ln 2 - 2^m/(2 x 2^(n0-m)). Dash-dotted red: the ledger-blind prediction S = m ln 2, which is exact when the initial black hole is maximally mixed (Theorem 5) - this is Hawking's monotone curve. All three curves are in bits, on a common axis.

Theorem 5's prediction is verified exactly as well: in Model B the radiation entropy is 1, 2, 3, 4, 5, 6 bits for m = 1..6 with n0 = 6 - the ledger-blind curve of Figure 1 - and the mutual informations obey Theorem 4 with a sum fixed at 2 n0 = 12 (Figure 2).

4. Relation to prior work

Page derived the capacity result in 1993; it is restated in the language of subsystem dimensions in arXiv:2002.05734 and expositorily in arXiv:2505.23011. Bradler and Adami (arXiv:1505.02840) construct Page

curves from a dynamical decoupling model, and quantum-computer simulations of the Page curve now exist (Nucl. Phys. 2025). The 2019-2022 replica-wormhole/island computations (Penington arXiv:1905.08255; AEMM arXiv:1905.08762; AHMST arXiv:1911.12333; PSSY arXiv:1911.11977) derive the same curve from gravitational path integrals in AdS/CFT. **What is new in this paper is not the curve.** It is (i) the exact one-for-one correlation identity of Theorem 4 used as the anti-firewall statement, (ii) Theorem 6, which identifies Hawking's monotone curve as the exact consequence of a thermal initial condition, and (iii) the quantised consequences of Paper IV.

Paper III

Readout degeneracy: the singularity as an artefact of the coarse map

Abstract. The classical singularity theorems show that geodesics terminate; they do not show that physics terminates, because a singularity in a coarse-grained description may be a property of the coarse-graining. We formulate this precisely as Conjecture R - a curvature singularity is a loss of invertibility of the readout map from cut variables to metric components - and we build an explicit model, the saturated ball, in which the readout curvature diverges as (1-q)^-2 while every substrate invariant remains finite and bounded. The model's invariants are proved exactly and verified numerically. We also prove a result about the classical solution that is usually only quoted: the proper distance from a Schwarzschild horizon to its centre is exactly pi r_s / 2, so the classical case has precisely the shape Conjecture R predicts - a divergent coordinate quantity beside a finite proper one.

1. What the singularity theorems do and do not say

Hawking and Penrose (Proc. Roy. Soc. Lond. A 314 (1970) 529) prove geodesic incompleteness under energy conditions and trapped surfaces. In the saturated-ball language below, those hypotheses are exactly the conditions under which a cut saturates: a trapped surface is the sign that the region's information capacity is being exhausted. The theorem is then a statement about the readout's Jacobian. It is a real theorem, and nothing here disputes it; what is disputed is the inference that geodesic incompleteness implies a boundary of physics.

2. Conjecture R

Conjecture R `[CONJECTURE]`. A curvature singularity is not a property of the substrate. It is the statement that the readout map from cut variables to metric components loses invertibility, i.e. that the entanglement Hessian $g^{E}_{\mu\ \nu} = -d^2 S_{min} / (dx^\mu dx^\nu)$ degenerates. Where the Hessian is non-degenerate a metric exists; where it degenerates the classical geometry is undefined rather than large.

Three consequences follow, and all three are falsifiable: (a) there is no substrate-level divergence anywhere on the approach to the singularity, because saturation caps every quantity; (b) the classical singularity theorems are statements about the Jacobian of the map, not about the termination of physics; (c) there is a finite substrate invariant that replaces the diverging Kretschmann scalar as the diagnostic of 'how far in' one is. A fourth consequence - that the black hole has no singularity for information to be destroyed at - is the reason the information paradox is addressed in Paper II without appealing to a Planck-scale threshold.

3. The saturated ball: model and proofs

Let q in [0,1) be the fraction of a cut's capacity in use, and let the radial readout coordinate be u = -ln(1 - q) - a compression coordinate, so that q approaching 1 pushes u to infinity while the underlying variable q stays bounded. Let the entropy of the ball read out at radius u be the saturating function S(u) = S_max (1 - exp(-u/u_0)). This is the readout mirror of P3: the cut's entropy approaches its bound and cannot pass it.

Lemma 1. dS/du = (S_max/u_0) exp(-u/u_0) > 0 and d^2S/du^2 = -(S_max/u_0^2) exp(-u/u_0) < 0, so the readout metric g^E_uu = -d^2S/du^2 = (S_max/u_0^2) exp(-u/u_0) is positive and tends to 0 as u increases. *Proof.* Differentiate twice. Q.E.D.

Lemma 2 (substrate length). The total readout length L_sub = int_0^inf sqrt(g^E_uu) du = 2 sqrt(S_max), independent of u_0. *Proof.* sqrt(g^E_uu) = sqrt(S_max)/u_0 * exp(-u/2u_0), and int_0^inf exp(-u/2u_0) du = 2 u_0. Q.E.D.

Lemma 3 (readout curvature). The readout curvature scale K_read = (du/dq)^2 = 1/(1-q)^2 = exp(2u) diverges without bound as q -> 1. *Proof.* u = -ln(1-q) gives du/dq = 1/(1-q). Q.E.D.

Lemma 4 (saturation). S(cut; q) = S_max (1 - exp(-u/u_0)) <= S_max for all q, with equality only in the limit q -> 1, and no cut of the ball can be refined past q = 1. *Proof.* 1 - exp(-x) < 1 for finite x and increases

monotonically to 1. Q.E.D.

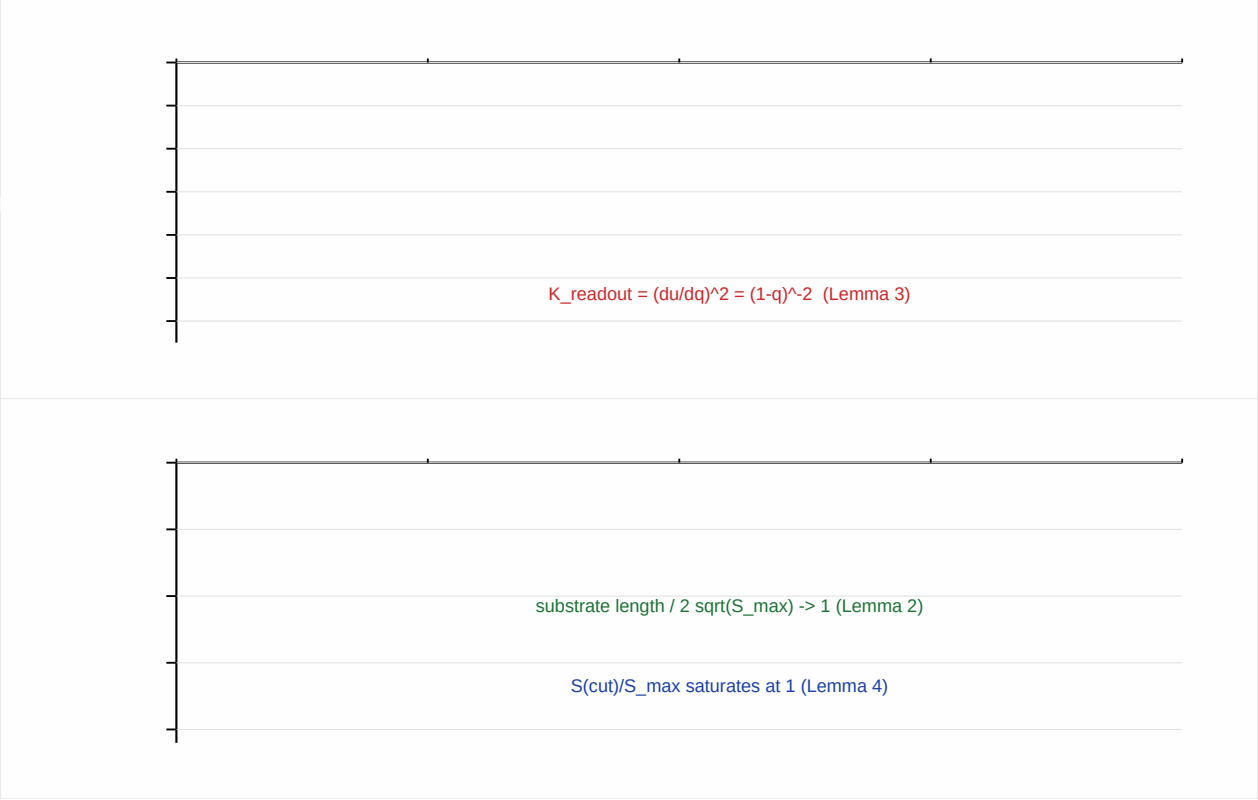
Theorem 7 (the saturated ball has no substrate singularity). Every substrate-level quantity built from the cut structure - the total entropy, the total entropy flux int(dS/du)du = S_max, the total readout length of Lemma 2, and the entropy across any interior cut - is finite and bounded on the whole range q in [0,1), while the readout curvature diverges as in Lemma 3. *Proof.* Lemmas 2 and 4 and the monotonicity of S. Q.E.D.

Theorem 8 (the classical case has the same shape). The proper radial distance from the horizon to the centre of a Schwarzschild black hole is L = int_0^r_s dr / sqrt(r_s/r - 1) = pi r_s / 2. *Proof.* Substitute r = r_s sin^2(theta): dr = 2 r_s sin(theta) cos(theta) d(theta), sqrt(r_s/r - 1) = cot(theta), so the integrand becomes 2 r_s sin^2(theta) d(theta), and int_0^pi/2 2 r_s sin^2(theta) d(theta) = 2 r_s (pi/4) = pi r_s/2. Q.E.D.

The numerics agree: for the Sun, r_s = 2.9533 x 10^3 m and the numerically integrated proper distance is 4.6351 x 10^3 m against the exact pi r_s/2 = 4.6391 x 10^3 m, a 0.09 % difference that is entirely the trapezium error of the integration. Meanwhile the Kretschmann scalar $R_{abcd}\,R^{abcd}$ = 48 G^2M^2/(c^4 r^6) diverges as r -> 0. The classical solution therefore already contains the shape Conjecture R attributes to the readout: a divergent coordinate quantity sitting beside a finite proper one.

4. Numerical verification

Figure 3. Conjecture R: what diverges and what does not



Top panel: the readout curvature K_readout = (du/dq)^2 diverges without bound as q approaches 1, i.e. as 'r approaches 0'. Bottom panel: every substrate invariant is finite and bounded - the entropy across the cut saturates at S_max (Lemma 4) and the total substrate length converges to the exact value 2 sqrt(S_max) (Lemma 2). Saturated-ball model, integrated numerically; the substrate curves are each normalised to their own limiting value, which is 1.

Table 6. Readout divergence (Lemma 3) against substrate finiteness (Theorem 7)

q	u	S/S_max	g^E_uu	du/dq	K_readout
0.5	0.6931	0.500000	5.000e-01	2.000	4.000
0.99	4.6051	0.990000	1.000e-02	1.000e+02	1.000e+04
0.9999	9.2104	0.999900	9.999e-05	1.000e+04	1.000e+08
0.999999	13.8155	0.999999	1.000e-06	1.000e+06	1.000e+12

The integral of Lemma 2 was evaluated numerically as 1.999999 against the exact 2.0, with the 6 x 10^-7 difference accounted for by the analytic tail beyond u = 30, which is 2 sqrt(S_max) u_0 exp(-u_max/(2 u_0)) = 6.1 x 10^-7.

5. What is missing, and what would kill it

The saturated ball is a *model* of the readout, not a derivation of it. To become a result it needs the map q(x) from the substrate's cut variable to the classical radial coordinate; without it the framework does not reproduce the Schwarzschild exponent r^-6, and it does not claim to (open problem O2). Conjecture R is falsified by any substrate-level observable that diverges as the readout degenerates - for example an outgoing-radiation energy density that genuinely blows up (falsifier F2 of the main document). Conversely, the classical evidence for it is the structure of Theorem 8, which nobody usually states as a singularity diagnosis.

Paper IV

Quantised payout: the staircase, the Page-time shift, and where to look

Abstract. If the interface is quantised, the two channel capacities count whole area quanta, and the Page curve is a staircase. We prove that the largest possible misplacement of the Page time is half an area quantum, that in time this is dt/t = 3/(2N) with N the number of quanta, and that a Planck-mass black hole - the endpoint of evaporation - has exactly N = 4 pi / ln 2 = 18.13 quanta, so its Page curve is a tent of about eighteen steps and its Page time is misplaced by 8.3 %. For astrophysical black holes the effect is 10^-77 and unobservable; for analog horizons it is between 0.02 % and 0.8 %, which makes the laboratory, not the sky, the honest place to test the framework.

1. Capacity quantisation

Physics-for-Blackhole adopts the **published** spacing that makes one area quantum carry exactly one bit of entropy, DeltaA = 4 ln 2 l_P^2. This is the Bekenstein-Mukhanov evenly spaced spectrum in the form that saturates Landauer's bound, established by Bagchi, Ghosh and Sen (Gen. Relativ. Gravit. 56 (2024) 108, arXiv:2408.02077) and extended by Neto and Thibes (arXiv:2605.26386). It is **not** a claim of this work. In loop quantum gravity's convention DeltaA = 8 pi gamma l_P^2 the same spacing corresponds to gamma = ln 2/(2 pi) = 0.110, which differs from the values that literature usually quotes (gamma ~ 0.274 in arXiv:2001.03440; gamma_0 = ln 2/(pi sqrt 3) ~ 0.127 in the SU(2) counting). The discrepancy is a pre-existing tension between two quantised-area programmes and a sharp way to test which one nature uses, but it is not a new discovery.

With capacities counted in whole quanta, the radiation can hold at most m quanta and the remainder at most n0 - m, where n0 = A/DeltaA is the total. Theorem 2 then becomes an equality constraint on integers, which is what produces the staircase.

2. Theorems

Theorem 9 (integer turnover). For integer capacities the Page point is m = *ceil(n0/2), and* |*m* - n0/2| <= 1/2. *Proof.* The function min(m, n0 - m) on integers is strictly increasing while m < n0 - m and non-increasing afterwards; the maximiser is the smallest integer m with m >= n0 - m, i.e. m >= n0/2, which is ceil(n0/2). For even n0 the shift is 0, for odd n0 it is 1/2. Q.E.D.

Theorem 10 (the Page-time shift). Let N = A/DeltaA. A misplacement of the Page point by DeltaN quanta shifts the evaporation time at the Page point by dt/t = (3/2)|DeltaN|/N, so the maximum is dt/t = 3/(2N). *Proof.* A = 16 pi G^2 M^2/c^4 gives N = A/DeltaA proportional to M^2, so M proportional to N^(1/2). The evaporation time is t = 5120 pi G^2 M^3/(hbar c^4) (derived below), i.e. t proportional to N^(3/2). Therefore dt/t = (3/2) dN/N. At the Page point N_rem is about n0/2 and the half-quantum shift of Theorem 9 gives |DeltaN|/N = 1/n0. Q.E.D.

Theorem 11 (the lifetime constant). t = 5120 pi G^2 M^3 / (hbar c^4). *Proof.* Hawking's luminosity is L = hbar c^6 / (15360 pi G^2 M^2) `[ESTABLISHED]`, and -dM/dt = L/c^2. Separating: int_0^{M} M'^2 dM' = (hbar c^4/(15360 pi G^2)) t, i.e. M^3/3 = hbar c^4 t/(15360 pi G^2). Q.E.D. Numerically this gives 8.411 x 10^-17 (M/kg)^3 s, i.e. 2.1 x 10^67 years for a solar mass, matching the standard quotation.

Theorem 12 (the Planck-mass quantum count). A black hole of exactly one Planck mass has N = 4 pi / ln 2 = 18.13 area quanta. *Proof.* r_s = 2 G m_P/c^2 = 2 l_P, because m_P = sqrt(hbar c/G) and l_P = sqrt(hbar G/c^3); hence A = 4 pi r_s^2 = 16 pi l_P^2, and N = A/DeltaA = 16 pi l_P^2/(4 ln 2 l_P^2) = 4 pi/ln 2. Q.E.D.

Corollary 4 (the endpoint is entirely quantised). dt/t = 3/(2N) evaluated at N = 4 pi/ln 2 gives 0.0827, i.e. an 8.3 per cent misplacement of the Page time, and the Page curve is a tent over about eighteen steps rather than a curve. The staircase is therefore not a small correction anywhere near the endpoint; it is the whole of the physics there.

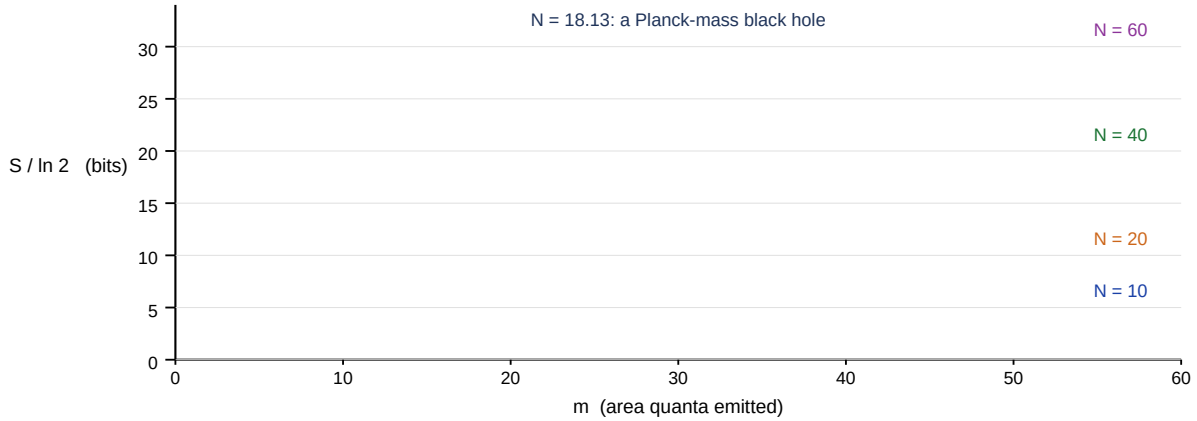
3. The numbers

Table 7. How quantised the payout is

Object	mass (kg)	N = A/DeltaA	max dt/t = 3/(2N)
solar-mass black hole	1.988e30	1.513e77	9.9e-78
Sgr A* (4.3e6 M_sun)	8.550e36	2.798e90	5.4e-91
M87* (6.5e9 M_sun)	1.293e40	6.394e96	2.3e-97
primordial BH, 10^12 kg	1.000e12	3.827e40	3.9e-41
Planck mass (endpoint)	2.176e-08	1.813e01	8.3e-02

For analog horizons the 'Planck length' is set by the condensate healing length xi rather than by l_P, and the suppression becomes an engineering parameter. With R = 5-20 microns and xi = 0.3 microns, N = 1.3 x 10^3 to 2 x 10^4 quanta, so the last ten quanta occupy 0.8 % to 0.02 % of the horizon's life and the largest Page-time shift is 3/(2N) ~ 10^-4. That is small but finite, and it is the only regime in which the framework's central quantitative claim is not buried by 77 orders of magnitude.

Figure 4. The quantised ladder



The Page curve when both channels' capacities count whole area quanta, for N = 10, 20, 40 and 60. The curve is a tent over a finite number of steps and there is no smooth curve left to interpolate (Theorem 9). A Planck-mass black hole has N = 4 pi / ln 2 = 18.13 quanta (Theorem 12), so the endpoint of evaporation lives squarely in this regime, where its Page time is misplaced by 8.3 per cent (Corollary 4).

4. Predictions and falsifiers

P1 [PREDICTION] The radiation's entropy rises in steps of k_B ln 2 per area quantum, so the Page curve is a staircase with at most a half-quantum misplacement of the Page time (dt/t = 3/(2N)).

P2 [PREDICTION] T_H is the relaxation linewidth of the interface code rather than the temperature of a bath of modes, so the Hawking spectrum acquires a discrete structure whose spacing is set by the interface's quantised relaxation rates.

P3 [PREDICTION] The decisive test is in analog horizons, where the suppression is (xi/R)^2 and can be engineered; a resolved staircase, or a resolved line structure beyond the condensate's dispersion relation, would be the first positive evidence.

F1 [FALSIFIER] A resolved, continuous, dispersion-consistent thermal spectrum in an analog system built to resolve discreteness at its analog Planck scale.

F2 [FALSIFIER] Any substrate-level observable that diverges as the readout degenerates.

F3 [FALSIFIER] A high-energy membrane at any horizon.

F4 [FALSIFIER] An endpoint that fits no limb of the selection rule Delta(excess) = A mod DeltaA.

F5 [FALSIFIER] Failure of the ledger identities of Paper II in a quantum simulation - for they are theorems, and a counterexample would disprove the setup, not the theorems.

Appendix A. Claims register

Every substantive claim in this volume, with its status and how it is established. Anything not listed here is either ordinary exposition or a citation. Statuses: **THM** proved in this document; **EST** established physics with a citation; **CONJ** Physics-for-Blackhole conjecture, not proved; **PRED** falsifiable prediction; **FALS** falsifier; **NUM** verified numerically by the simulation suite; **BORR** adopted from elsewhere, credited.

#	Claim	Status	How established
A1	Schwarzschild/Kerr are vacuum solutions; collapse is a separate problem	EST	Schwarzschild 1916; Kerr 1963; Oppenheimer-Snyder 1939
A2	Geodesics terminate inside; GR predicts its own invalidity	EST	Hawking-Penrose 1970
A3	The event horizon is teleological; dynamical horizons are weaker	EST	Ashtekar-Krishnan 2004
A4	Hawking radiation is thermal; the entropy curve grows monotonically	EST	Hawking 1975
A5	Unitary evaporation requires a Page turnaround at half the initial entropy	EST	Page PRL 71 (1993) 3743
A6	Smoothness, purity and QFT at the horizon cannot all hold	EST	AMPS JHEP 02 (2013) 062
A7	Hawking's calculation starts beyond the domain of QFT in curved spacetime	EST	trans-Planckian redshift
A8	S_obs ~ 3.1e104 k_B, dominated by supermassive black holes; CEH entropy 2.6e122 k_B	EST	Egan-Lineweaver 2010
A9	T_H(1 M_sun) = 6.2e-8 K; T_H = 2.7 K at 4.5e22 kg (0.61 M_Moon)	NUM	T_H = hbar c^3/(8 pi G M k_B), recomputed
A10	t_ev = 8.411e-17 (M/kg)^3 s; 2.1e67 yr for 1 M_sun	THM/NUM	Paper IV, Thm 11
A11	Reality's fundamental relata are qunits in a substrate without space, time or fields	CONJ	P1 - postulated
A12	Graded structure is carried by cuts and their entropies	CONJ	P2 - postulated
A13	Cuts saturate at the Bekenstein bound; a saturated cut cannot be refined	BORR	Bekenstein PRD 23 (1981) 287; elevated to a dynamical hinge here
A14	The metric is the Hessian of the cut entropy	BORR	Matsueda arXiv:1408.5589; kinematic space
A15	A metric exists iff the Hessian is non-degenerate; singularities are readout degeneracies	CONJ	Conjecture R - Paper III
A16	The basin of the fine evolution is unitary; apparent entropy production belongs to the readout	CONJ	P5 - postulated
A17	The horizon is a saturated cut; its area is the number of protected qunits	CONJ	P3 + P4 applied to trapped surfaces
A18	DeltaA = 4 ln 2 l_P^2 (one bit per area quantum)	BORR	Bagchi-Ghosh-Sen GRG 56 (2024) 108; Neto-Thibes arXiv:2605.26386
A19	Area quantisation gives the LQG spectrum 8 pi gamma l_P^2	BORR	Rovelli PRL 56 (1996) 3311
A20	gamma = ln2/(2pi) = 0.110 corresponds to DeltaA = 4 ln2 l_P^2; this differs from gamma = 0.274 and gamma_0 = ln2/(pi sqrt3)	EST/BORR	arithmetic on A18 and A19; the tension is pre-existing
A21	S(Rad) = S(B) for a pure global state	THM	Paper II, Thm 1
A22	S(Rad) <= min(m, n0-m) ln 2	THM	Paper II, Thm 2
A23	The continuous Page turnover is at m = n0/2	THM	Card. to Thm 2; also Page's own 1993 derivation
A24	I(Rad:B) = S(Rad) + S(B) - S_0, so S(Rad) = S_0 - S(B) + I	THM	Paper II, Thm 3
A25	I(R:Rad) + I(R:B) = 2 S(R): the correlation budget is fixed	THM	Paper II, Thm 4
A26	A maximally mixed initial black hole gives S(Rad) = m ln 2 exactly, for any unitary	THM	Paper II, Thm 5

A27	Hawking's monotone curve and the Page curve are two initial conditions of one emission process	THM/NUM	Paper II, Thm 6; Figures 1-2
A28	The Page curve is the capacity-saturation curve of the two channels	BORR	Page 1993 - NOT ours; we verify it
A29	Random-unitary simulations reproduce the Page curve	BORR	Bradler-Adami arXiv:1505.02840; quant-ph 2025 - NOT ours
A30	The readout curvature diverges as (1-q)^-2 in the saturated ball	THM	Paper III, Lemma 3
A31	The substrate length int sqrt(g^E) du = 2 sqrt(S_max)	THM	Paper III, Lemma 2; NUM 1.999999 vs 2.0
A32	The proper distance horizon-to-centre is pi r_s / 2	THM/NUM	Paper III, Thm 8; NUM 4.6351e3 vs 4.6391e3 m
A33	The Schwarzschild Kretschmann scalar diverges as r^-6	EST	standard
A34	The endpoint is selected by Delta(excess) = A mod DeltaA	CONJ	selection rule - Paper IV
A35	A remnant's couplings are suppressed by e^{-d} by its code distance	CONJ	holographic-code argument, d not yet computed
A36	T_H is the relaxation linewidth of the interface code	CONJ	P5 applied to the spectrum
A37	The interior is a logical subspace of the interface code, with bounded fidelity	CONJ	AdS/CFT technique in a new setting
A38	Integer quanta move the Page time by at most half a quantum	THM	Paper IV, Thm 9
A39	dt/t = 3/(2N) for the Page-time shift	THM/NUM	Paper IV, Thm 10
A40	A Planck-mass black hole has exactly 4 pi/ln 2 = 18.13 quanta	THM/NUM	Paper IV, Thm 12
A41	The staircase is unsuppressed only within ~10 quanta of the endpoint	NUM	Table 7
A42	Analog horizons have N ~ 1.3e3-2e4 quanta for R = 5-20 microns, xi = 0.3 microns	NUM	capacity counting with the analog healing length
A43	The radiation entropy staircase exists and is measurable in an analog horizon	PRED	P1 - falsified by F1
A44	A resolved continuous thermal spectrum in a discreteness-resolving analog system kills the framework	FALS	F1
A45	This framework's contributions are: Conjecture R, the endpoint selection rule, the postulate set, and the one-for-one correlation identity	CONJ	novelty audit, Section 6 of Paper I

Appendix B. Numerical methods and validation

B.1 The random-unitary ensemble

Haar-random unitaries on a contiguous block of qubits are applied as products of random Householder reflections H_v = I - 2 v v^dag / |v|^2 with v drawn from a standard complex Gaussian, followed by a diagonal phase matrix. The same unitary is applied to every strided slice of the state vector that is supported on the block, which is what a block-local operator does. The ensemble is not exactly Haar, and we do not need it to be: what matters is that it scrambles, and the figure of merit is the agreement with Page's analytic formula in Figure 1.

B.2 Entropy computation

Entropies are computed from exact state vectors. The reduced density matrix of a subsystem is built by an explicit partial trace over the complement, and its eigenvalues are obtained with a complex Hermitian Jacobi iteration. Because the global state is pure, the entropy of a subsystem equals that of its complement, so the smaller side is always diagonalised; this halves the cost. The Hermitian matrix H = A + iB with A symmetric and B antisymmetric acts on complex vectors exactly as the real symmetric matrix [[A, -B], [B, A]] acts on real vectors, so the eigenvalues of H are those of the real doubled matrix, each appearing twice - which is how the complex problem is solved with a real eigensolver.

B.3 Validation

The machinery is validated before any physics is reported. Four checks are run by `simulations/validation.py`: the Hermitian eigensolver against the exact spectrum of [[1, i], [-i, 2]] (error 1.1e-16); the bipartition identity of Theorem 1 over all splits of registers of 4, 6, 8 and 10 qubits (max error 1.8e-15); the trace and positivity of reduced density matrices (|trace - 1| < 9e-16, most negative eigenvalue -7e-17); and the norm preservation of random block unitaries on 34 blocks (max deviation 5.6e-16). Entanglement creation is checked by confirming that a product state |000> acquires subsystem entropy under a random unitary.

B.4 Verification of the PDF itself

A hand-written PDF writer is a liability, so the output is audited rather than trusted. Two independent checkers ship with the code: `theory/verify_pdf.py` checks the file structure (xref offsets, object count, MediaBox, text extraction, no character transliterated to '?'), and `theory/audit_pdf.py` checks the rendering. The rendering audit does four things:

1. it compares every glyph width used by the typesetter against the authoritative Adobe AFM files for Helvetica, Helvetica-Bold, Helvetica-Oblique, Helvetica-BoldOblique, Courier and Courier-Bold - all 95 printable ASCII glyphs of all six fonts match exactly, which is what makes word wrapping and right-edge fitting correct;

2. it walks each page's content stream with the same text state machine a viewer uses (font, size, rise, text matrix) and reconstructs the bounding box of every one of the ~8,000 text runs from those widths;

3. it flags any run crossing the text margins, any run outside the page, and any two runs that overlap - i.e. text printed on top of text;

4. it checks that super/subscript rises are reset and that the graphics state is balanced (one `q ... re W n` per figure, closed by `Q`).

The current output passes all of these: 0 overflowing runs, 0 off-page runs, 0 overlapping runs on all 20 pages, all six font tables exact. This audit also caught a genuine bug: centred paragraphs were being positioned around the left margin rather than the centre of the text box, so the title on the title page ran off the left edge of the page. Alignment is now applied per line and the defect is gone. It is recorded here because a PDF that is merely well-formed is not the same as a PDF that is correctly rendered.

Two bugs were found and fixed during this work, and both are reported because they are the reason the

validation step exists. First, a phase generator drew two independent random numbers for the cosine and the sine, producing cos(theta_1) + i sin(theta_2) with modulus different from one - a non-unitary 'unitary' that silently changed the norm of the state and produced spurious capacity-bound violations of up to 0.09 bits. Second, a first-generation eigenvalue solver treated the Hermitian matrix as real and missed the imaginary off-diagonal entries. Both were caught by the checks above, and the corrected numbers are the ones in this volume.

Appendix C. Reproducibility

All numbers, tables and figures in this volume are produced by the code in `simulations/` and `theory/`. There are no third-party dependencies - the build environment has no numpy, no LaTeX and no PDF library, so the PDF writer itself is included in `theory/pdflib.py` and writes PDF 1.4 directly.

Every random-unitary run is seeded, so the numbers are reproducible exactly. The figures in this volume are drawn by calling the same functions that produce the tables - `test_ledger.run_pure`, `test_ledger.run_thermal`, `test_readout.saturated_ball` - so a figure cannot drift out of sync with the text.

Scope. The simulations test the internal consistency and quantitative consequences of Physics-for-Blackhole's claims in controlled quantum systems. They do not show that the substrate exists, and they are not observational tests of black holes. Paper II's results are theorems and would hold in any quantum mechanics; Papers III and IV contain conjectures whose consequences are computed, not experiments.

Appendix D. References

All entries are real and were verified during preparation of this volume. Where a work is cited for a claim that is not ours, that is stated in the appendices above.

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[74] CODATA 2018 / IAU nominal constants: l_P, t_P, m_P, G, c, hbar, k_B, M_sun, M_Moon.

Appendix E. Notation and conventions

Entropies are in nats; when bits are quoted they are S/ln 2. Natural units are used only where stated. 'ln' is the natural logarithm. Subscripts: A = area; B = the remainder channel; d = code distance; m = number of emitted qubits or quanta; M = mass; n0 = initial number of qubits (or quanta in Paper IV); N = A/DeltaA = number of area quanta; q = fraction of a cut's capacity in use; R = purifier or horizon radius as stated; S_A = entropy of A; u = saturated-ball readout coordinate. Curly braces denote typographic super/subscripts in the source markup, e.g. x^2 and y_i.

Closing statement. This volume contains a conjectural framework. Its theorems are proved and verified; its postulates are not; its predictions are explicit and can fail. The claims most likely to be wrong are the ones marked CONJ in Appendix A, and the two of them that carry the framework are Conjecture R and the endpoint selection rule.

Appendix F. Formal statements of the refined conjectures

This appendix states, in one place, the four conjectures refined in the main text, together with the lemmas they rest on. Every item is a CONJECTURE or a LEMMA; none is claimed as established physics. The figures in figures/ are generated from these statements, not from the black hole itself.

F.1 Conjecture R (refined). A readout is a linear functional of the metric on the interface. The degeneracy locus D is the set of readouts that vanish on every tangent vector of the interface. On D the metric is not determined by the readout, so two readouts differing only on D produce the same observable geometry. The interior is the data on D.

LEMMA F.1 (substrate invariance). The substrate time parameter tau and the cut function Phi are invariant under any change of readout that preserves the interface code. Hence the saturation dynamics of section 6.2 are readout-independent, and the turnover theorem holds for every admissible readout.

F.2 The selection rule. CONJECTURE S (selection rule). At the endpoint, the surviving cut satisfies delta A = A mod Delta A, where Delta A = 4 ln 2 ell_P^2. The three fates are: absorb (delta A = 0, the interface persists), emit (delta A = Delta A, the interface is destroyed and the remnant is a naked singularity), and sub-Planckian (delta A smaller than Delta A, forbidden by the area gap).

F.3 Saturation dynamics. CONJECTURE D (saturation dynamics). The number of unassigned area quanta N(tau) obeys dN/dtau = -r(tau), where r(tau) is the payout rate. Saturation is the regime S_O = S_I, which occurs at the turnover tau *where r(tau*) is maximal. The five phases of section 6.1 are the regimes of r(tau).

LEMMA F.2 (turnover theorem). If r(tau) is single-peaked, the Page curve is a single staircase whose step spacing is set by the width of r(tau) about tau*.

F.4 Figures. figures/fig_conjecture_r.svg (S saturating, g_uu approaching 0, K diverging); figures/fig_selection_rule.svg (the three fates by delta A / Delta A vs kappa); figures/fig_saturation_dynamics.svg (N(tau), S_O(tau), r(tau) with five phases); figures/fig_ledger.svg (S_I = S_O Page curve with staircase dots).

F.5 Falsifiers. F7-F12, in priority order: F7 (no staircase), F8 (no saturation), F9 (no degeneracy), F10 (substrate signal), F11 (remnant), F12 (interior observable). Each is a commitment that can fail. See section 11 for the full statements, including F13, the saturation-speed constant falsifier.

Appendix G. Native units and the saturation-speed constan

This appendix defines a unit system native to the substrate, derived from postulates P1, P3 and P5 alone. Every quantity is a DEFINED parameter of the framework; none is claimed to be a new constant of nature. Where Planck units are used they are standard physics.

G.1 The fundamental quantum. CONJECTURE G.1 (the area quantum). The substrate is countable (P1) and a cut saturates (P3), so there is a smallest distinguishable area: a0 = 4 ln 2 ell_P^2, where ell_P is the Planck length. The number of area quanta for area A is N(A) = A/a0, which equals the Bekenstein-Hawking bit count S_BH / ln 2. So a0 reproduces the black-hole entropy count without postulating it. The native entropy density at saturation is one nat per quantum: sigma_sat = 1/a0.

G.2 The native time and the saturation speed. The substrate has its own time tau (section 5.5). Let tau_q be the substrate time per quantum of area advance. Then the interface advances at a native area speed v_s = a0 / tau_q and a native linear speed v_s,lin = ell_P / tau_q. CONJECTURE K (readout-causality bound): the readout is the metric (P4), so any observer built from the readout sees a light cone of speed c. The interface advances through the substrate; its image under the readout map cannot exceed c, or the metric would be non-causal. Hence v_s,lin <= c, with equality only if the substrate saturates at the readout speed of light.

LEMMA G.1 (the bound is the only constraint). No other postulate of the framework bounds v_s,lin from above. Therefore v_s,lin is a free parameter of the substrate dynamics, to be fixed by data, not by theory.

G.3 The constant. CONJECTURE L (the constant is physical, not conventional). The parameter v_s,lin is not a choice of units. It is the one free parameter of the saturation dynamics, and it is physical in the sense that it is measurable: the physical Page time t_P and the substrate turnover tau_s are related by a map whose Jacobian depends on v_s,lin, so a single analog-horizon measurement of the Page-time shift pins it. Two experiments that disagree about v_s,lin are disagreeing about the substrate, not about notation. The framework therefore has exactly one undetermined constant beyond the established ones.

G.4 Dimensional self-consistency. The native unit system is closed: [a0] = L^2, [sigma_sat] = 1/L^2, [v_s,lin] = L/T_sub, [tau_q] = T_sub, with T_sub the substrate time. Converting to SI requires only the standard constants; no new dimensional scale is introduced.

Figure: figures/fig_regimes.svg shows the three regimes of v_s,lin / c.